COMPLETE EXPERIMENTAL REPORT

Experiments 17 and 18 | CEBRA / GRU | Mihili CO (2014)

Neural Z-score ablation, standard PGD, Noise-Aware training, and Exp17 fine epsilon sweep

Report scope: dataset and training protocol, 24 primary Exp17 runs, 24 Exp18 runs, 12 additional Exp17 fine-sweep runs, per-day results, and direct comparisons. This report is based on the run outputs shared in the project and the corresponding runner source code.

1. Dataset and shared experimental protocol

Component	Specification
Animal / task	Mihili monkey performing the center-out (CO) reaching task
Dataset identifier	mihili_co / Mihili_CO_2014_raw
Sessions	2014 dataset; 11 recording sessions/days indexed Day 0 through Day 10
Neural inputs	96 neural channels; spike counts in 50-ms bins; xds_smooth=False
Labels	Continuous behavioral targets used by the CEBRA time_delta model and decoder evaluation
Training partition	The first 80% of Day 0 only; both encoder and decoder trained on this partition
Within-day evaluation	The last 20% of Day 0 (Day 0 holdout R^2)
Cross-day evaluation	The last 20% of each target day (Days 1–10); no encoder or decoder retraining on target days
Primary metric	Mean R^2 across Days 1–10, then averaged over random seeds 42 and 43
Architecture provenance	Independent version derived from monkey_gru; Git commit a7808902cf721ea8b440684f48fea7514da8c881

Time-based splitting uses shuffle=False. These results cover the Mihili CO dataset only; they do not include Jango, Chewie, or Mihili RT.

2. Model architecture and common training settings

Setting	Value / description
Encoder	MonkeyEncoder with Offset36_multi; CEBRA time_delta; 36-bin input context, 64-dimensional latent representation, hidden size 256
Decoder	MonkeyDecoder based on a unidirectional GRU; hidden size 512, two layers, dropout 0.4, 256-bin window
Objective / stopping	CEBRA contrastive learning for the encoder; behavioral prediction loss for decoder; early stopping as implemented in the original code
Batch size	1,024
Maximum optimization steps	Encoder: 30,000; decoder: 30,000 (potentially fewer with early stopping)
Optimizer / learning rate	Adam; encoder LR linearly decays from 0.02 to 0.002 using LinearLR
Additional encoder settings	weight_decay=1e-5; adam_eps=0.1; gradient clipping=5.0
Encoder perturbation	L2 PGD with the implemented per-bin perturbation convention; 10 attack steps
Decoder perturbation	Enabled for ALL experimental arms: adv_dec=True, epsilon=0.1, 10 L2 steps
Random seeds	42 and 43, with independently trained models for each arm and seed
Compute environment	EPFL RunAI, H200 GPUs; Exp18 and the fine sweep used four independent GPU workers

Important labeling detail: “Clean Encoder” means that adversarial training of the ENCODER is disabled. The decoder is still adversarially trained with ε=0.1 even in the Clean arm.

3. Controlled difference between Exp17 and Exp18

Condition	Exact setup
Exp17: Z-score enabled	For each day d, channelwise means and standard deviations were computed using the first 80% of that same day; that day’s data were then standardized using these statistics. Channels with standard deviation < 0.001 were set to zero.
Exp18: Z-score disabled	Neural activity from all days was fed to the model as original 50-ms binned spike counts, without Z-scoring. Behavioral target labels were unchanged.
Held constant	Animal, sessions, architecture, 80/20 temporal splits, seeds, optimization schedule, experimental arms, and R^2 evaluation.
Noise-Aware calibration	For Exp18, the Noise-Aware covariance and AUTO epsilon values were recomputed from raw Day 0 training data only. The internal whitening geometry of Noise-Aware was not removed.

Interpretation caveat: Exp17 uses statistics from the initial 80% of each TARGET day during preprocessing, whereas Exp18 does not. A numeric ε has different physical meaning on standardized versus raw counts. Accordingly, this is a preprocessing ablation, not a comparison at identical physical perturbation magnitude.

4. Compared methods: 12 arms per experiment

Arm	Definition
Clean Encoder	No adversarial encoder attack; decoder adversarial training remains enabled (ε=0.1).
Standard PGD	Original implementation with Euclidean L2 perturbation and ε=0.5.
Noise-Aware: manual	Whitened per-bin perturbation budgets ε = 0.1, 0.3, 0.5, 0.7, 1, 3, 5, and 7.
AUTO Empirical	Automatically selects ε using Day 0 training data and empirical multiplier 0.363; decoder-independent.
AUTO Quantile	Automatically selects ε from a calibrated lower (5th) quantile of behavioral distances in Day 0 training data; decoder-independent.

Noise-Aware estimates a covariance matrix of neural-activity differences (with Ledoit–Wolf shrinkage) to define the perturbation geometry: $\delta_t = \Sigma^{(1/2)} z_t$ with $||z_t||_2 \leq \varepsilon$ per bin. Standard PGD projects perturbations in input-space Euclidean geometry. Consequently, the epsilon values from the two families are not directly comparable.

AUTO calibration used 20,000 sampled pairs from Day 0 training data only. The empirical rule uses data-derived ε_beh scaled by 0.363, while the quantile rule uses the 5th percentile of calibrated behavioral distances over a 36-bin context, subject to the implemented channel-dependent cap.

5. Exp17 main results: with per-day Z-score

All 24 primary Exp17 runs (12 experimental arms × two seeds) completed. The main metric is mean cross-day R^2 over Days 1–10; Day 0 holdout is excluded from this mean.

Method	Seed 42	Seed 43	Day 0	Mean D1–D10
Standard PGD, ε=0.5	0.6663	0.6820	0.6382	0.6742
Noise-Aware, ε=0.1	0.6589	0.6712	0.6607	0.6650
Noise-Aware, ε=0.3	0.6227	0.6450	0.6339	0.6338
Noise-Aware, ε=0.5	0.6273	0.5993	0.6167	0.6133
Clean Encoder	0.6039	0.5903	0.6354	0.5971
Noise-Aware, ε=0.7	0.5960	0.5515	0.5780	0.5737
Noise-Aware, ε=1	0.5449	0.5419	0.5041	0.5434
AUTO Quantile	0.5393	0.5028	0.4208	0.5211
AUTO Empirical	0.5213	0.5147	0.4644	0.5180
Noise-Aware, ε=5	0.4873	0.4831	0.6164	0.4852
Noise-Aware, ε=7	0.4360	0.4875	0.5528	0.4617
Noise-Aware, ε=3	0.3577	0.4309	0.5885	0.3943

6. Exp18 main results: without Z-score

All 24 Exp18 models were trained independently of Exp17. “Positive cases” counts the instances with $R^2 > 0$ among 20 cross-day evaluations (10 days × two seeds). The positive-only mean averages only successful cases and is not a replacement for the primary mean.

Method	Seed 42	Seed 43	Mean D1–D10	Positive	Pos. mean
Standard PGD, ε=0.5	0.4223	0.4048	0.4135	20/20	0.4135
AUTO Quantile	0.2095	0.2968	0.2531	18/20	0.2875
AUTO Empirical	0.2076	0.2322	0.2199	17/20	0.2732
Noise-Aware, ε=0.3	0.1168	0.2161	0.1665	13/20	0.3328
Noise-Aware, ε=1	0.1323	0.1959	0.1641	12/20	0.3438
Noise-Aware, ε=3	0.2465	0.0658	0.1562	15/20	0.2525
Noise-Aware, ε=0.7	0.1398	0.1656	0.1527	14/20	0.2762
Noise-Aware, ε=0.5	0.0898	0.1343	0.1120	14/20	0.2427
Noise-Aware, ε=0.1	-0.0172	0.1800	0.0814	12/20	0.2899
Clean Encoder	-0.1860	-0.1779	-0.1820	8/20	0.1208
Noise-Aware, ε=5	-1.3889	0.0528	-0.6680	6/20	0.1426
Noise-Aware, ε=7	-1.0721	-0.8892	-0.9807	0/20	—

7. Exp17 supplemental fine sweep: smaller epsilon

Only the Noise-Aware encoder-training arm was evaluated on the Z-scored Exp17 setup. Six new epsilon values were trained with both seeds (12 additional runs), while results for ε=0.1 were reused from the primary Exp17 experiment rather than retrained.

Epsilon	Seed 42	Seed 43	Mean R^2	Run provenance
0.01	0.6358	0.5763	0.6060	New fine-sweep run
0.025	0.6446	0.6394	0.6420	New fine-sweep run
0.05	0.6725	0.6519	0.6622	New fine-sweep run
0.075	0.6746	0.6533	0.6640	New fine-sweep run
0.1	0.6589	0.6712	0.6651	From primary Exp17
0.15	0.6561	0.6602	0.6582	New fine-sweep run
0.2	0.6485	0.6408	0.6446	New fine-sweep run

Within the measured fine-sweep points, ε=0.1 yielded the highest mean cross-day R^2 (about 0.6650). The means at ε=0.075 and ε=0.05 were 0.6640 and 0.6622, respectively. Standard PGD at ε=0.5 achieved 0.6742 in the primary Exp17 experiment.

8. AUTO-selected perturbation budgets

Experiment	Method	Selected ε	Mean cross-day R^2
Exp17 (Z-score)	AUTO Empirical	1.378016	0.5180
Exp17 (Z-score)	AUTO Quantile	1.454291	0.5211
Exp18 (raw)	AUTO Empirical	1.699369	0.2199
Exp18 (raw)	AUTO Quantile	1.962792	0.2531

Each selected epsilon was identical across seeds 42 and 43 within a given experiment because calibration used the same Day 0 training data. Epsilon was not selected using decoder hyperparameters or cross-day test R^2.

9. Direct comparison: Exp17 versus Exp18

Method	Exp17 (Z-score)	Exp18 (raw)	Difference: 18 − 17
Standard PGD, ε=0.5	0.6742	0.4135	-0.2607
Noise-Aware, ε=0.1	0.6650	0.0814	-0.5836
Noise-Aware, ε=0.3	0.6338	0.1665	-0.4673
Noise-Aware, ε=0.5	0.6133	0.1120	-0.5013
Clean Encoder	0.5971	-0.1820	-0.7791
Noise-Aware, ε=0.7	0.5737	0.1527	-0.4210
Noise-Aware, ε=1	0.5434	0.1641	-0.3793
AUTO Quantile	0.5211	0.2531	-0.2680
AUTO Empirical	0.5180	0.2199	-0.2981
Noise-Aware, ε=5	0.4852	-0.6680	-1.1532
Noise-Aware, ε=7	0.4617	-0.9807	-1.4424
Noise-Aware, ε=3	0.3943	0.1562	-0.2381

Across all 12 methods, average cross-day R^2 decreased when the per-day Z-score step was removed. Clean Encoder dropped from 0.5971 to −0.1820; standard PGD dropped from 0.6742 to 0.4135. Better within-day Day 0 R^2 in Exp18 does not imply better transfer to later days.

10. Exp17 per-day results: average of both seeds

Each value is averaged across seeds 42 and 43. D0 is the holdout split of the training day; the “Mean” column averages D1–D10 only.

Method	D0	D1	D2	D3	D4	D5
Standard PGD, ε=0.5	0.6382	0.6823	0.7202	0.6769	0.7067	0.7092
Noise-Aware, ε=0.1	0.6607	0.7399	0.5932	0.7056	0.7193	0.7397
Noise-Aware, ε=0.3	0.6339	0.7031	0.5301	0.6526	0.7057	0.7315
Noise-Aware, ε=0.5	0.6167	0.6734	0.4981	0.6008	0.6743	0.7149
Clean Encoder	0.6354	0.6945	0.5224	0.6575	0.6557	0.6687
Noise-Aware, ε=0.7	0.5780	0.6233	0.4720	0.5515	0.6482	0.7016
Noise-Aware, ε=1	0.5041	0.5621	0.4441	0.5315	0.6337	0.6772
AUTO Quantile	0.4208	0.5443	0.4600	0.4949	0.6207	0.6598
AUTO Empirical	0.4644	0.5370	0.4124	0.5113	0.5981	0.6834
Noise-Aware, ε=5	0.6164	0.2311	0.5749	0.5525	0.3988	0.5655
Noise-Aware, ε=7	0.5528	0.3732	0.4590	0.5279	0.3307	0.6042
Noise-Aware, ε=3	0.5885	0.4798	0.4179	0.3500	0.2239	0.2895

Method	D6	D7	D8	D9	D10	Mean
Standard PGD, ε=0.5	0.6395	0.7602	0.6895	0.4751	0.6821	0.6742
Noise-Aware, ε=0.1	0.6592	0.7919	0.7109	0.3951	0.5955	0.6650
Noise-Aware, ε=0.3	0.6397	0.7723	0.6828	0.3607	0.5596	0.6338
Noise-Aware, ε=0.5	0.6342	0.7731	0.6555	0.3912	0.5174	0.6133
Clean Encoder	0.5561	0.7338	0.6202	0.3456	0.5165	0.5971
Noise-Aware, ε=0.7	0.6189	0.7482	0.6312	0.2662	0.4762	0.5737
Noise-Aware, ε=1	0.5953	0.7373	0.6105	0.1985	0.4441	0.5434
AUTO Quantile	0.5684	0.7140	0.5959	0.1726	0.3801	0.5211
AUTO Empirical	0.5714	0.7187	0.6093	0.1397	0.3983	0.5180
Noise-Aware, ε=5	0.4772	0.6964	0.5026	0.3452	0.5081	0.4852
Noise-Aware, ε=7	0.4758	0.6227	0.4592	0.2214	0.5433	0.4617
Noise-Aware, ε=3	0.4769	0.6729	0.3531	0.1955	0.4831	0.3943

11. Exp18 per-day results: average of both seeds

Per-day R^2 values for all 12 Exp18 methods are shown without clipping negative scores. All values reflect the raw-input/no-Z-score condition.

Method	D0	D1	D2	D3	D4	D5
Standard PGD, ε=0.5	0.6885	0.6417	0.7316	0.5756	0.1640	0.1207
AUTO Quantile	0.7359	0.0980	0.4541	0.1673	0.0829	0.1364
AUTO Empirical	0.7389	-0.0093	0.4447	0.0535	-0.1053	0.1598
Noise-Aware, ε=0.3	0.7710	0.1527	0.4519	-0.0459	-0.2208	0.2964
Noise-Aware, ε=1	0.7526	0.0127	0.4476	0.0041	-0.1943	0.1046
Noise-Aware, ε=3	0.6116	0.0371	0.3714	0.2176	-0.0904	0.1356
Noise-Aware, ε=0.7	0.7538	0.0574	0.4672	0.0418	-0.2336	0.0687
Noise-Aware, ε=0.5	0.7615	0.0934	0.4825	0.0059	-0.2144	0.1045
Noise-Aware, ε=0.1	0.7762	0.1416	0.4636	-0.0531	-0.3039	0.0804
Clean Encoder	0.7393	0.0771	0.3556	-0.1647	-0.4619	-0.2185
Noise-Aware, ε=5	0.3590	-0.6393	-0.2244	-0.3860	-0.7582	-0.7715
Noise-Aware, ε=7	0.4339	-0.9099	-0.4893	-0.8831	-1.1048	-1.0451

Method	D6	D7	D8	D9	D10	Mean
Standard PGD, ε=0.5	0.1659	0.5010	0.4099	0.3779	0.4473	0.4135
AUTO Quantile	0.0477	0.4834	0.0819	0.3546	0.6252	0.2531
AUTO Empirical	0.1036	0.3759	0.1330	0.4367	0.6067	0.2199
Noise-Aware, ε=0.3	-0.2180	0.2628	0.0316	0.4093	0.5453	0.1665
Noise-Aware, ε=1	-0.1856	0.3467	0.0077	0.4396	0.6582	0.1641
Noise-Aware, ε=3	-0.0690	0.2595	-0.0402	0.2592	0.4808	0.1562
Noise-Aware, ε=0.7	-0.1472	0.3729	0.0248	0.3400	0.5348	0.1527
Noise-Aware, ε=0.5	-0.3437	0.2178	0.0102	0.3040	0.4600	0.1120
Noise-Aware, ε=0.1	-0.4937	0.2499	0.0155	0.2576	0.4560	0.0814
Clean Encoder	-0.7815	0.0044	-0.4984	-0.1782	0.0461	-0.1820
Noise-Aware, ε=5	-0.8125	-0.5731	-0.8378	-1.0034	-0.6741	-0.6680
Noise-Aware, ε=7	-1.0802	-0.9247	-1.1230	-1.2850	-0.9616	-0.9807

12. Key numerical outcomes and limitations

• Exp17 (Z-score): Standard PGD, ε=0.5, had the highest cross-day mean R^2 (0.6742). The best measured Noise-Aware setting in the fine sweep was ε=0.1, with R^2=0.6650.

• Exp18 (no Z-score): Standard PGD, ε=0.5, remained best (R^2=0.4135). AUTO Quantile ranked second (R^2=0.2531) and achieved positive R^2 in 18 of 20 cross-day evaluations.

• Effect of preprocessing: The mean cross-day R^2 declined by approximately 0.7791 for Clean Encoder and 0.2607 for standard PGD when moving from Exp17 to Exp18 (differences based on rounded output values).

• Scope of the claim: These R^2 scores were measured on held-out recordings without an artificial attack at test time. They do not directly quantify test-time adversarial robustness. Perturbation budgets differ in geometry and scale between standard PGD and Noise-Aware, and between standardized and raw inputs.

13. Result storage and reproducibility

Item	Location or file
Exp17 source	/home/mirzaei/sam/result/Aggregate/EXP17_MIHILI_CO/source
Exp17 results	/mnt/upmwmathis/scratch/hossein/mm_project/EXP17_MIHILI_CO/runs
Fine-sweep results	/mnt/upmwmathis/scratch/hossein/mm_project/EXP17_FINE_SWEEP/runs
Exp18 source	/home/mirzaei/sam/result/Aggregate/EXP18_MIHILI_CO/source
Exp18 results	/mnt/upmwmathis/scratch/hossein/mm_project/EXP18_MIHILI_CO/runs
Primary artifacts	metrics.json and exp17_record.json or exp18_record.json for each run

Evidence base: final RunAI logs, complete Exp18 result text from November 9, 2026, Exp17 tables and fine-sweep outputs provided in the conversation, and the Exp17/Exp18 runner source archives. More than two random seeds and evaluation under test-time attacks would be needed for stronger statistical or robustness claims.